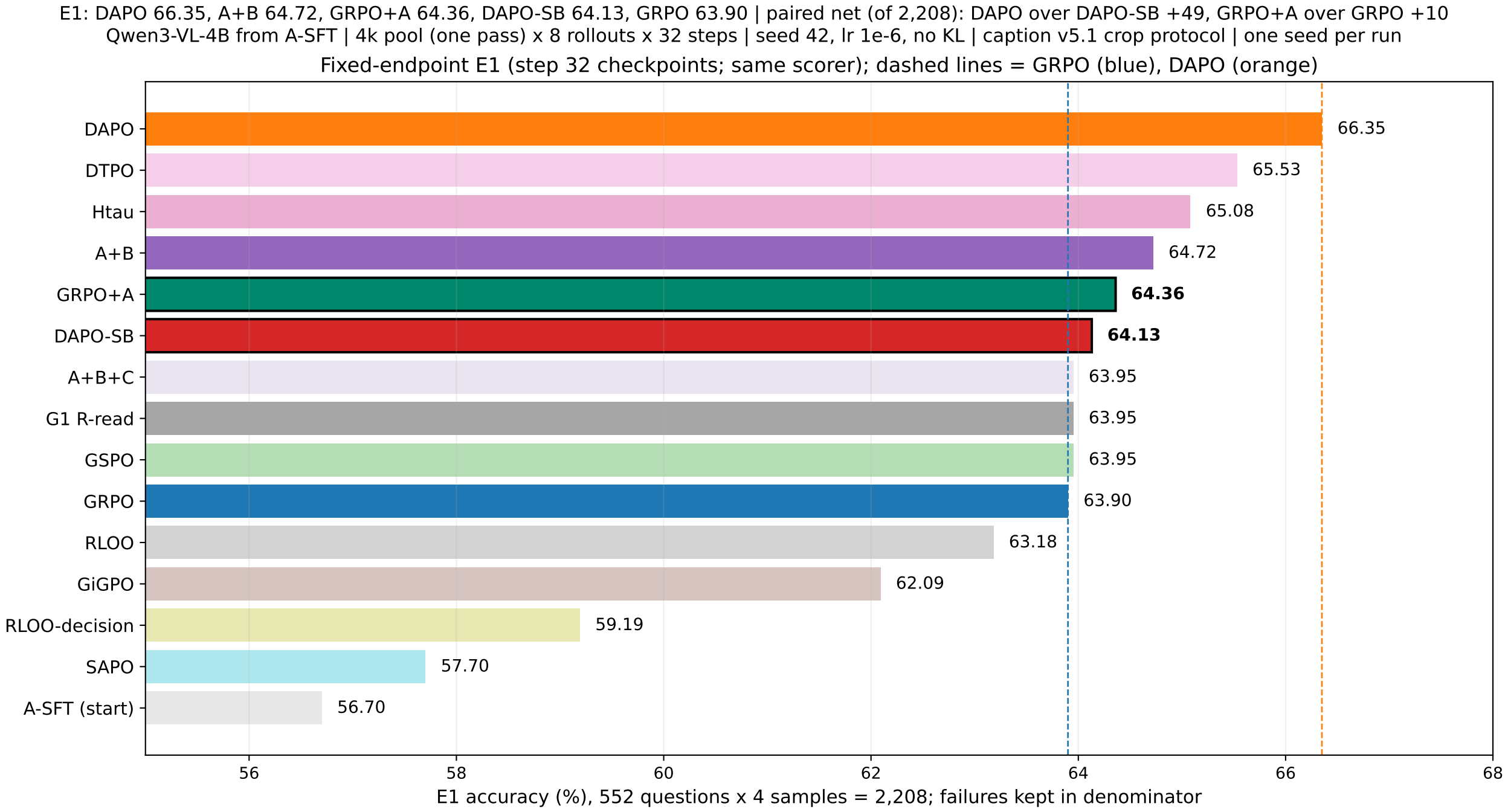
E1: DAPO 66.35, A+B 64.72, GRPO+A 64.36, DAPO-SB 64.13, GRPO 63.90 | paired net (of 2,208): DAPO over DAPO-SB +49, GRPO+A over GRPO +10
Qwen3-VL-4B from A-SFT | 4k pool (one pass) x 8 rollouts x 32 steps | seed 42, lr 1e-6, no KL | caption v5.1 crop protocol | one seed per run
Fixed-endpoint E1 (step 32 checkpoints; same scorer); dashed lines = GRPO (blue), DAPO (orange)
DAPO
DTPO
Htau
A+B
GRPO+A
DAPO-SB
A+B+C
G1 R-read
GSPO
GRPO
RLOO
GiGPO
RLOO-decision
SAPO
A-SFT (start)
66.35
65.53
65.08
64.72
64.36
64.13
63.95
63.95
63.95
63.90
63.18
62.09
59.19
57.70
56.70
56
58
60
62
64
66
68
E1 accuracy (%), 552 questions x 4 samples = 2,208; failures kept in denominator